



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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SECTION : DBSE&DBD Y25 Section-8

MySQL Views

MYSQL VIEWS

Practical Examples and Student Exercises using MySQL Workbench

1. What is a View?

A **View** is a **virtual table** created from one or more existing tables.

- It **does not store data** itself.
- It stores only the **SQL query**.
- Whenever you query the view, MySQL retrieves the latest data from the underlying tables.

2. Setup: Employees and Departments Tables

Run this setup first. If these tables already exist with data, you may skip the setup.

```
CREATE DATABASE IF NOT EXISTS employeedb1;
```

```
USE employeedb1;
```

```
CREATE TABLE IF NOT EXISTS departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE IF NOT EXISTS employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    emp_salary INT,  
    emp_dept_id INT  
);
```


MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

msql x

Limit to 1000 rows

```

266 e.salary
267 FROM employees e
268 JOIN departments d
269 ON e.dept_id = d.dept_id
270 WHERE d.dept_name = 'CSE';
271
272 • SELECT * FROM CSEEmployees;

```

Result Grid

emp_id	emp_name	dept_name	salary
101	Rahul	CSE	50000.00
103	Anil	CSE	55000.00

employees 87 employees 88 EmployeeView 89 EmployeeBasicView 90 HighSalaryEmployees 91 EmployeeDepartmentView 92 CSEEmployees 93 Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
387	15:22:46	DROP VIEW IF EXISTS EmployeeDepartmentView	0 row(s) affected	0.000 sec
388	15:22:46	CREATE VIEW EmployeeDepartmentView AS SELECT e.emp_id, ...	0 row(s) affected	0.016 sec
389	15:22:46	SELECT * FROM EmployeeDepartmentView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
390	15:22:46	DROP VIEW IF EXISTS CSEEmployees	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeedb1.cse...	0.000 sec
391	15:22:46	CREATE VIEW CSEEmployees AS SELECT e.emp_id, e.emp_na...	0 row(s) affected	0.000 sec
392	15:22:46	SELECT * FROM CSEEmployees LIMIT 0, 1000	2 row(s) returned	0.016 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

msql x

Limit to 1000 rows

```

251     d.dept_name,
252     e.salary
253 FROM employees e
254 JOIN departments d
255 ON e.dept_id = d.dept_id;
256
257 • SELECT * FROM EmployeeDepartmentView;

```

Result Grid

emp_id	emp_name	dept_name	salary
101	Rahul	CSE	50000.00
102	Priya	ECE	65000.00
103	Anil	CSE	55000.00
104	Sneha	EEE	70000.00
105	Kiran	ECE	48000.00

68 Result 69 employees 70 employees 71 EmployeeView 72 EmployeeBasicView 73 HighSalaryEmployees 74 EmployeeDepartmentView Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 320	15:22:10	DROP VIEW IF EXISTS HighSalaryEmployees	0 row(s) affected	0.000 sec
✓ 321	15:22:10	CREATE VIEW HighSalaryEmployees AS SELECT emp_id, emp_name, ...	0 row(s) affected	0.000 sec
✓ 322	15:22:10	SELECT * FROM HighSalaryEmployees LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
⚠ 323	15:22:10	DROP VIEW IF EXISTS EmployeeDepartmentView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeedb1.empl...	0.000 sec
✓ 324	15:22:10	CREATE VIEW EmployeeDepartmentView AS SELECT e.emp_id, ...	0 row(s) affected	0.015 sec
✓ 325	15:22:10	SELECT * FROM EmployeeDepartmentView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

mysql x

Limit to 1000 rows

```
237
238 • CREATE VIEW HighSalaryEmployees AS
239     SELECT emp_id, emp_name, salary
240     FROM employees
241     WHERE salary > 55000;
242
243 • SELECT * FROM HighSalaryEmployees;
```

Result Grid

emp_id	emp_name	salary
102	Priya	65000.00
104	Sneha	70000.00

Result Grid

Form Editor

result 50 Result 51 Result 52 Result 53 employees 54 employees 55 EmployeeView 56 EmployeeBasicView 57 HighSalaryEmployees Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 251	15:21:26	DROP VIEW IF EXISTS EmployeeBasicView	0 row(s) affected	0.000 sec
✓ 252	15:21:26	CREATE VIEW EmployeeBasicView AS SELECT emp_id, emp_name, s...	0 row(s) affected	0.000 sec
✓ 253	15:21:26	SELECT * FROM EmployeeBasicView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
⚠ 254	15:21:26	DROP VIEW IF EXISTS HighSalaryEmployees	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeb1.high...	0.000 sec
✓ 255	15:21:26	CREATE VIEW HighSalaryEmployees AS SELECT emp_id, emp_name, ...	0 row(s) affected	0.016 sec
✓ 256	15:21:26	SELECT * FROM HighSalaryEmployees LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

msql x

Limit to 1000 rows

```
228 • DROP VIEW IF EXISTS EmployeeBasicView;
229
230 • CREATE VIEW EmployeeBasicView AS
231 SELECT emp_id, emp_name, salary
232 FROM employees;
233
234 • SELECT * FROM EmployeeBasicView;
```

Result Grid

	emp_id	emp_name	salary
▶	101	Rahul	50000.00
	102	Priya	65000.00
	103	Anil	55000.00
	104	Sneha	70000.00
	105	Kiran	48000.00

Form Editor

employees 34 Result 35 Result 36 Result 37 Result 38 employees 39 employees 40 EmployeeView 41 EmployeeBasicView Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 185	15:20:49	DROP VIEW IF EXISTS EmployeeView	0 row(s) affected	0.000 sec
✓ 186	15:20:49	CREATE VIEW EmployeeView AS SELECT * FROM employees	0 row(s) affected	0.000 sec
✓ 187	15:20:49	SELECT * FROM EmployeeView LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
⚠ 188	15:20:49	DROP VIEW IF EXISTS EmployeeBasicView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeedb1.empl...	0.000 sec
✓ 189	15:20:49	CREATE VIEW EmployeeBasicView AS SELECT emp_id, emp_name, s...	0 row(s) affected	0.000 sec
✓ 190	15:20:49	SELECT * FROM EmployeeBasicView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

mysql x

Limit to 1000 rows

```
220 • DROP VIEW IF EXISTS EmployeeView;
221
222 • CREATE VIEW EmployeeView AS
223 SELECT *
224 FROM employees;
225
226 • SELECT * FROM EmployeeView;
```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

Export: Wrap Cell Content: [F12](#)

Result Grid

Form Editor

17 Result 18 Result 19 employees 20 Result 21 Result 22 Result 23 Result 24 employees 25 employees 26 EmployeeView Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 122	15:20:03	UPDATE employees SET salary = 70000 WHERE emp_id = 104	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
✓ 123	15:20:03	UPDATE employees SET salary = 48000 WHERE emp_id = 105	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
✓ 124	15:20:03	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
⚠ 125	15:20:04	DROP VIEW IF EXISTS EmployeeView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeeb1.empl...	0.000 sec
✓ 126	15:20:04	CREATE VIEW EmployeeView AS SELECT * FROM employees	0 row(s) affected	0.016 sec
✓ 127	15:20:04	SELECT * FROM EmployeeView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

mysql x

Limit to 1000 rows

```
212 • UPDATE employees SET salary = 50000 WHERE emp_id = 101;
213 • UPDATE employees SET salary = 65000 WHERE emp_id = 102;
214 • UPDATE employees SET salary = 55000 WHERE emp_id = 103;
215 • UPDATE employees SET salary = 70000 WHERE emp_id = 104;
216 • UPDATE employees SET salary = 48000 WHERE emp_id = 105;
217
218 • SELECT * FROM employees;
```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00
NULL	NULL	NULL	NULL

Result 3 employees 4 Result 5 Result 6 employees 7 Result 8 Result 9 Result 10 Result 11 employees 12 employees Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
62	15:19:21	UPDATE employees SET salary = 50000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
63	15:19:21	UPDATE employees SET salary = 65000 WHERE emp_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
64	15:19:21	UPDATE employees SET salary = 55000 WHERE emp_id = 103	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
65	15:19:21	UPDATE employees SET salary = 70000 WHERE emp_id = 104	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
66	15:19:21	UPDATE employees SET salary = 48000 WHERE emp_id = 105	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
67	15:19:21	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

msql x

Limit to 1000 rows

```

264     AVG(e.salary) AS AverageSalary
265 FROM employees e
266 JOIN departments d
267 ON e.dept_id = d.dept_id
268 GROUP BY d.dept_id, d.dept_name;
269
270 • SELECT * FROM DepartmentSalaryView;

```

Result Grid

dept_name	EmployeeCount	AverageSalary
CSE	2	52500.000000
ECE	2	56500.000000
EEE	1	70000.000000

employees 107 EmployeeView 108 EmployeeBasicView 109 HighSalaryEmployees 110 EmployeeDepartmentView 111 DepartmentSalaryView 1: Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 460	15:28:53	DROP VIEW IF EXISTS EmployeeDepartmentView	0 row(s) affected	0.016 sec
✓ 461	15:28:53	CREATE VIEW EmployeeDepartmentView AS SELECT e.emp_id, ...	0 row(s) affected	0.000 sec
✓ 462	15:28:53	SELECT * FROM EmployeeDepartmentView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
⚠ 463	15:28:53	DROP VIEW IF EXISTS DepartmentSalaryView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employee_db1.dep...	0.000 sec
✓ 464	15:28:53	CREATE VIEW DepartmentSalaryView AS SELECT d.dept_name, ...	0 row(s) affected	0.000 sec
✓ 465	15:28:53	SELECT * FROM DepartmentSalaryView LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

mysql x

Limit to 1000 rows

```

277 FROM employees e
278 JOIN departments d
279 ON e.dept_id = d.dept_id
280 GROUP BY d.dept_id, d.dept_name
281 HAVING AVG(e.salary) > 50000;
282
283 • SELECT * FROM HighAverageDepartments;

```

Result Grid

dept_name	AverageSalary
ECE	56500.000000
EEE	70000.000000

Form Editor

View 126 EmployeeBasicView 127 HighSalaryEmployees 128 EmployeeDepartmentView 129 DepartmentSalaryView 130 HighAverageDepartments 1 Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 535	15:29:45	DROP VIEW IF EXISTS DepartmentSalaryView	0 row(s) affected	0.000 sec
✓ 536	15:29:45	CREATE VIEW DepartmentSalaryView AS SELECT d.dept_name, ...	0 row(s) affected	0.015 sec
✓ 537	15:29:45	SELECT * FROM DepartmentSalaryView LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
⚠ 538	15:29:45	DROP VIEW IF EXISTS HighAverageDepartments	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employee...'	0.000 sec
✓ 539	15:29:45	CREATE VIEW HighAverageDepartments AS SELECT d.dept_name...	0 row(s) affected	0.000 sec
✓ 540	15:29:45	SELECT * FROM HighAverageDepartments LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local MySQL x

File Edit View Query Database Server Tools Scripting Help

mysql x

Limit to 1000 rows

```

284 • DROP VIEW IF EXISTS EmployeeCountView;
285
286 • CREATE VIEW EmployeeCountView AS
287 SELECT COUNT(*) AS TotalEmployees
288 FROM employees;
289
290 • SELECT * FROM EmployeeCountView;

```

Result Grid

TotalEmployees
5

View 146 HighSalaryEmployees 147 EmployeeDepartmentView 148 DepartmentSalaryView 149 HighAverageDepartments 150 EmployeeCountView 1 Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 613	15:30:27	DROP VIEW IF EXISTS HighAverageDepartments	0 row(s) affected	0.000 sec
✓ 614	15:30:27	CREATE VIEW HighAverageDepartments AS SELECT d.dept_name...	0 row(s) affected	0.000 sec
✓ 615	15:30:27	SELECT * FROM HighAverageDepartments LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
⚠ 616	15:30:27	DROP VIEW IF EXISTS EmployeeCountView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employee...'	0.000 sec
✓ 617	15:30:27	CREATE VIEW EmployeeCountView AS SELECT COUNT(*) AS TotalE...	0 row(s) affected	0.016 sec
✓ 618	15:30:27	SELECT * FROM EmployeeCountView LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Next view - 9

MySQL Workbench

Local Mysql x

File Edit View Query Database Server Tools Scripting Help

msql x

Limit to 1000 rows

```

294 SELECT
295     emp_id,
296     emp_name,
297     salary
298 FROM employees;
299
300 • SELECT * FROM EmployeeSalaryView;

```

Result Grid

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

Export: Wrap Cell Content: Form Editor

Employees 167 EmployeeDepartmentView 168 DepartmentSalaryView 169 HighAverageDepartments 170 EmployeeCountView 171 EmployeeSalaryView 172 Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
694	15:31:35	DROP VIEW IF EXISTS EmployeeCountView	0 row(s) affected	0.000 sec
695	15:31:35	CREATE VIEW EmployeeCountView AS SELECT COUNT(*) AS TotalE...	0 row(s) affected	0.016 sec
696	15:31:35	SELECT * FROM EmployeeCountView LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
697	15:31:35	DROP VIEW IF EXISTS EmployeeSalaryView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeedb1.empl...	0.016 sec
698	15:31:35	CREATE VIEW EmployeeSalaryView AS SELECT emp_id, emp_n...	0 row(s) affected	0.000 sec
699	15:31:35	SELECT * FROM EmployeeSalaryView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local Mysql x

File Edit View Query Database Server Tools Scripting Help

msql x

Limit to 1000 rows

```

294 SELECT
295     emp_id,
296     emp_name,
297     salary
298 FROM employees;
299
300 • SELECT * FROM EmployeeSalaryView;

```

Result Grid

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

Export: Wrap Cell Content: Form Editor

Employees 167 EmployeeDepartmentView 168 DepartmentSalaryView 169 HighAverageDepartments 170 EmployeeCountView 171 EmployeeSalaryView 172 Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
694	15:31:35	DROP VIEW IF EXISTS EmployeeCountView	0 row(s) affected	0.000 sec
695	15:31:35	CREATE VIEW EmployeeCountView AS SELECT COUNT(*) AS TotalE...	0 row(s) affected	0.016 sec
696	15:31:35	SELECT * FROM EmployeeCountView LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
697	15:31:35	DROP VIEW IF EXISTS EmployeeSalaryView	0 row(s) affected, 1 warning(s): 1051 Unknown table 'employeeedb1.empl...	0.016 sec
698	15:31:35	CREATE VIEW EmployeeSalaryView AS SELECT emp_id, emp_n...	0 row(s) affected	0.000 sec
699	15:31:35	SELECT * FROM EmployeeSalaryView LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec